Egocentric Human Demonstrations for Bimanual Garment Manipulation:

From Hand-Worn-Gripper Recordings to Robot Policies

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Technical report, status of 5 November 2026
Repository: `egocentric-garment-manipulation`

***Abstract*—We want a two-arm robot to manipulate a T-shirt using human demonstrations recorded with hand-worn grippers, without teleoperation and without robot data collection. The demonstrator wears two HandUMI grippers (an open-source design by BrikHMP18, not ours) with fiducial markers, a wrist fisheye camera, an IMU and, since 29 October 2026, the jaw servo read as a torque-off encoder; a fixed RGB-D camera observes both hands. All poses are expressed in one world frame, the robot base, set by a camera-to-robot calibration (0.98 px held-out), start markers measured in that frame and a calibrated marker rig (1–4 mm static error), and all streams share one host clock. The jaw-tip pose comes from point tracking on the fixed camera (CoTracker3) anchored by the markers, with the IMU only inside tracking gaps: on held-out takes the error at hidden markers has a median of 16 mm (left) and 19 mm (right), against about 42 mm for markers with IMU fill alone; wrist-camera SLAM was tried and rejected because it loses tracking exactly at grasps. As an end-to-end test of the data, converted demonstrations were first replayed open loop on two reBot B601-DM arms: four replays of one 33 s demonstration repeat within about 1 mm, the jaws close 5–19 mm above the table (human about 10 mm), and the robot grasped and dragged the shirt, but did not fold it. We then built policy training data: 300 successful T-shirt flattening demonstrations, graded per demo, with labels taken *before* inverse kinematics and without any clamp. First fine-tuned policies, observed in robot trials that were not counted, stop short of the cloth and close the jaws early. Two causes were recorded (the policy copied the jaw state from its input; the robot wrist cameras sit differently from the hand-worn ones), and checking the labels found three more data problems (a label clamp, demonstration speed, video jaw labels). In a reduced feasibility test, pinching one sleeve with the left hand, a policy fine-tuned on 197 demonstrations grasped the sleeve in real-robot trials on 2 November 2026; the trials were not counted, and after a miss the policy grasps elsewhere instead of retrying. Back on flattening, a policy trained with the cloth-folding recipe of UMI on 250 demonstrations under a re-aimed camera improves its offline score from epoch to epoch, and a vision classifier now labels whether a closed jaw holds the cloth, because the jaw servo has no load signal. We report no success rate.**

I. Introduction

Folding a T-shirt is long-horizon, bimanual and contact-rich, and the cloth state changes with every grasp. Teleoperated robot data for such tasks is slow and expensive to collect. Hand-held and hand-worn grippers in the style of UMI [1] let a person demonstrate with an end-effector that resembles the robot's, without a robot in the loop, so data collection scales with people rather than robots. The difficulty moves into the conversion: every error in calibration, pose recovery, jaw labelling, timing or kinematic feasibility ends up in the training labels, and a policy trained on those labels inherits it. Policy success rates are a late and indirect signal for such errors.

Main question. *If a robot simply replays the converted human motion, open loop, does it do what the human did?* Replay exercises every stage at once (calibration, pose, jaw state, robotization, IK, timing, control), puts human, robot command and measured robot motion on one clock and in one frame so that errors can be attributed to a stage, and fails visibly. The follow-up question is whether the same conversion yields labels from which a policy can learn a grasp.

Contributions of the work so far:

- one world frame, the robot base, set once by calibration and start markers, so that human and robot paths share it without per-take alignment (Sec. V);
- a pose source for hand-worn grippers that survives the grasp: fixed-camera point tracking anchored by fiducial markers, with the IMU only in gaps, validated at held-out markers (Sec. VI); jaw labels from a servo encoder (Sec. VII);
- an open-loop replay test in which human, command and robot are compared on one clock, with its measured results (Sec. VIII);
- a training-data pipeline with per-demo grades and Cartesian labels before IK, the first policies on the robot, the recorded causes of their failure and the data problems found afterwards (Secs. IX–X).

Status (2 November 2026). Replay is finished; folding by replay does not succeed. Policies fine-tuned on the flattening data were run on the robot in uncounted trials and did not grasp. In a one-sleeve pinching test (Sec. XI), successful grasps were observed in real-robot trials on 2 November; the trials were not counted. No success rate is reported anywhere in this report, because no counted trials were run.

Relation to earlier versions. This report replaces version 0.1 (27 October) and version 0 (21 October), which described the replay track in full detail, and the short technical note; they are kept in `report/archive/`. Replay-era details that no longer affect the current pipeline were removed here; superseded methods are summarised in Table VIII.



Fig. 1. The setup seen by the fixed RGB-D camera: a T-shirt on the table, the demonstrator's two hand-worn grippers with fiducial markers resting on the two start markers, and the two robot arms at the left and right edges.

II. Related Work

Hand-held and hand-worn grippers. UMI [1] records demonstrations with a hand-held parallel-jaw gripper and a wrist fisheye camera; its pose comes from monocular-inertial SLAM based on ORB-SLAM3 [21], and trajectories are filtered by the kinematics of the target robot. HandUMI [2], [3], designed and published by BrikHMP18 and co-authors, is an open-source hand-worn variant for robot-free bimanual data capture; we use a build copy of it. DexUMI [4] uses the human hand with an exoskeleton as the interface for dexterous hands. We keep the UMI-style gripper but take the pose from a fixed RGB-D camera instead of wrist SLAM, and we test the data on the target robot before policy training. Instead of removing infeasible trajectories, we keep the human path as the label and leave feasibility to the executor.

Human data and low-cost bimanual robots. EgoMimic [5] co-trains on egocentric human video and robot data. ALOHA [6] showed fine-grained bimanual manipulation with low-cost arms and teleoperation; we use two low-cost reBot B601-DM arms but record without teleoperation, through LeRobot [7].

Cloth. Garment folding and unfolding have been learned with dedicated primitives, e.g. SpeedFolding [8] and FlingBot [9]. We do not propose a cloth method; our policies are generic diffusion-transformer [26] and VLA [28] policies.

Pose, calibration, kinematics. Markers are AprilTags [10], [11] detected with OpenCV [12], solved with IPPE [13] and SQPnP [14]. Point tracking uses CoTracker3 [15]; rotation uses a Mahony filter [16]. The camera-to-robot solve is initialised with the method of Shah [17]. IK uses Pink [18] on Pinocchio [19]. Evaluated and rejected as pose sources: MASt3R-SLAM [20], ORB-SLAM3, and with an earlier stereo camera Basalt [22], OpenVINS [23] and AirSLAM [24].

III. System Overview

Figure 2 shows the current data flow. Two principles run through it. *One world frame*: every pose, human or robot, is expressed in the robot base frame; no per-recording alignment (no SLAM map, no point-cloud registration) is needed. *One clock*: camera frames, IMU and servo samples and the robot's commanded and measured joints all carry host timestamps, so human, command and robot can be compared sample by sample and filmed side by side by the same camera without synchronisation hardware.

TABLE I
Components

unit	model, role
Hand-worn gripper ×2	HandUMI [2] by BrikHMP18 (not our design), our build; demonstration tool
Markers	three 48 mm AprilTags per gripper, since 23 October a fourth 30 mm tag on the back of the wrist camera
Wrist camera ×2	fisheye, 1280×720 at 30 Hz; jaws and local cloth view
Wrist IMU ×2	Seeed XIAO nRF52840 Sense; gyro and accelerometer
Jaw servo ×2	Feetech STS3215, torque off, read at about 100 Hz on the host clock (removed 13–28 October, back since 29 October)
RGB-D camera	Orbbec Gemini 336, colour + aligned depth (recorded at about 15 Hz), fixed above the table; raised to 113 cm on 25 October (height not re-measured after the angle-only re-aim of 1 November)
Robot arms ×2	reBot B601-DM, 6-DoF + parallel jaw, Damiao motors on USB-CAN, our MIT-mode bridge at 100 Hz
Host	Apple-silicon laptop: recording, IK, bridge, deploy loop
GPU	rented RTX 5090 pods for point tracking, the jaw model, the 3-D solve and some policy training; one run on a rented H100; RTX 4090 via Ray for the DiT fine-tunes

IV. Hardware

Table I lists the components.

HandUMI is not our design. It was designed and published by BrikHMP18 (murobotics-ai, https://github.com/murobotics-ai/handumi-hw, Apache-2.0) [2], [3] and follows UMI [1]. We print the upstream parts (commit `e58de33`) and made these changes (Fig. 3): a wrist IMU instead of VR tracking; a modified controller-support connector, camera mount and servo cover; several AprilTags per hand on different faces, so that at least one tends to face the fixed camera, plus since 23 October a small tag on the back of the wrist camera, whose single-tag planar pose flip is resolved with the IMU.

Jaw servo timeline. The jaw servos were removed on 13 October, because the servo shafts broke and the units were heavy; from 13 to 28 October the jaw state was inferred from the wrist video. On 29 October the servo was put back, now with torque off so that it never drives the jaw while a person holds it; it is only read, as an encoder of the jaw opening (Sec. VII).

Robot. Two reBot B601-DM arms (the vendor LeRobot plugin and URDF are not included in the repository) are driven by our HTTP bridge in MIT mode: one 100 Hz thread,

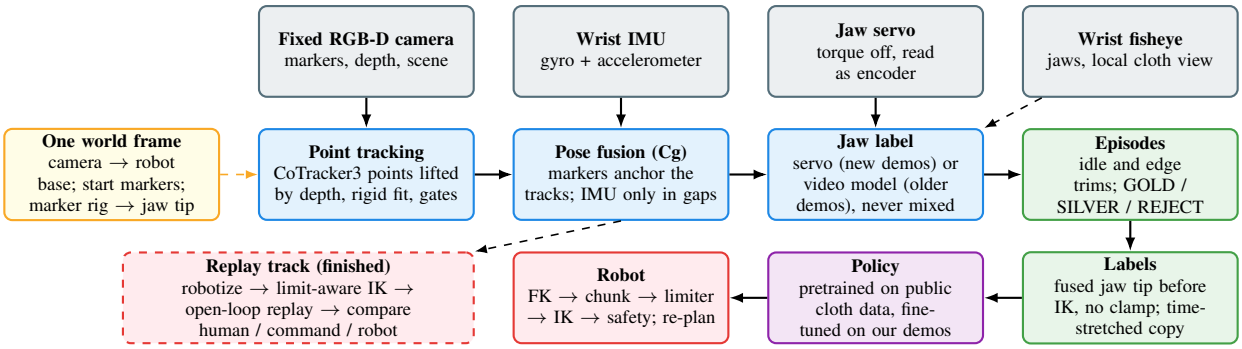


Fig. 2. Data flow as of 1 November 2026. Grey: sensors, all stamped by one host clock. Yellow: one-time setup that puts human and robot in one world frame. Blue: pose and jaw state of each gripper. Green: graded episodes and training labels. Purple and red: policy and execution on the robot. Dashed box: the open-loop replay track, which tested the conversion before policy training (Sec. VIII). The wrist video is a policy input; it supplies jaw labels only for demonstrations recorded before the servo was reinstalled.

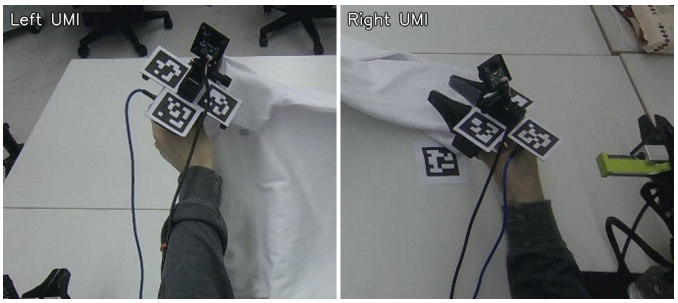


Fig. 3. The two hand-worn grippers as recorded before 23 October: HandUMI (BrikHMP18's design) with three 48 mm AprilTags per hand, wrist fisheye and IMU; no jaw servo in that period. Since 23 October a fourth, 30 mm tag sits on the back of each wrist camera, and since 29 October the jaw servo is back as an encoder.

TABLE II
CALIBRATION RESULTS

item	result
Camera ↔ robot base	0.98 px held-out reprojection, ≈1.3 mm lateral
Corrected robot model, absolute jaw position	within 6–9 mm of tape at the home pose
Start markers 400 / 401	$(225, 166, -4)$ mm and $(243, -194, 6)$ mm
Marker rig → jaw tip	1–4 mm static; 1.8–4.8 mm leave-one-out; 5–7 mm RMS on an independent pivot test
Human ↔ robot gripper orientation	spread 2–9°

acceleration-limited motion toward each goal, goal interpolation between streamed commands, automatic re-enable of a disabled motor, USB reconnect, and an enforced jaw-opening limit. The bridge uses an integral-only feed-forward torque and no gravity model; Sec. VIII explains why this determined the tracking error.

V. One World Frame and Calibration

Human and robot data use one world frame B, the robot base frame ($+x$ forward, $+y$ left, $+z$ up). The jaw-tip TCP is the midpoint between the jaw tips, defined the same way for the human gripper and the robot gripper. $T_{XY} \in SE(3)$ maps coordinates in frame Y to frame X. The frame is set up in four steps:

1) *Camera to robot.* First calibration: the robot holds a marker on its wrist in 32 poses in front of the fixed camera; the unknowns T_{CB} and the marker-to-TCP transform satisfy $T_{C,m}(k) = T_{CB}\, T_{BE}(q_k)\, T_{E,m}$, initialised with Shah's method [17] and refined on the reprojection of all tag corners plus a depth residual (18 of the 32 poses used; five-fold held-out validation). The camera was moved for the flattening takes (raised 96→105→113 cm); after every move T_{CB} is re-solved without moving the robot: two fixed 70 mm table tags with known robot coordinates fix position and yaw, and the table plane in the depth image fixes tilt and height (`robot_cam_v002-v005`).
2) *Start markers.* The two start markers are measured once with the calibrated camera (Fig. 4a, Table II).
3) *Marker rig to jaw tip.* Slow rotation takes give the tag-to-tag transforms of each gripper's rig; the jaw-tip offset t_{tcp} in the rig frame comes from touching the corners of a board of known geometry, $t_{\mathrm{tcp}} = \arg\min_t \sum_j \rho(\|p_{B,\mathrm{rig}}(j) + R_{B,\mathrm{rig}}(j)\, t - c_B(j)\|)$ with a Huber loss ρ, and a pivot take gives an independent check.
4) *Start of every take.* Each take starts with the grippers at the start markers (Fig. 4b).

Robot model corrections measured in the world. Three errors of the robot model were found only by measuring the real robot with tape and camera: the URDF gripper TCP lies 89 mm in front of the real jaw tips; the left arm base is 70 mm inward of the URDF with a 3° tilt; and the left arm's second joint has a −8.4° zero offset. All three are applied as overrides inside the IK model (`configs/laundry/robot/`); the URDF file is untouched. They were invisible to FK-based parity, which compares the model with itself.

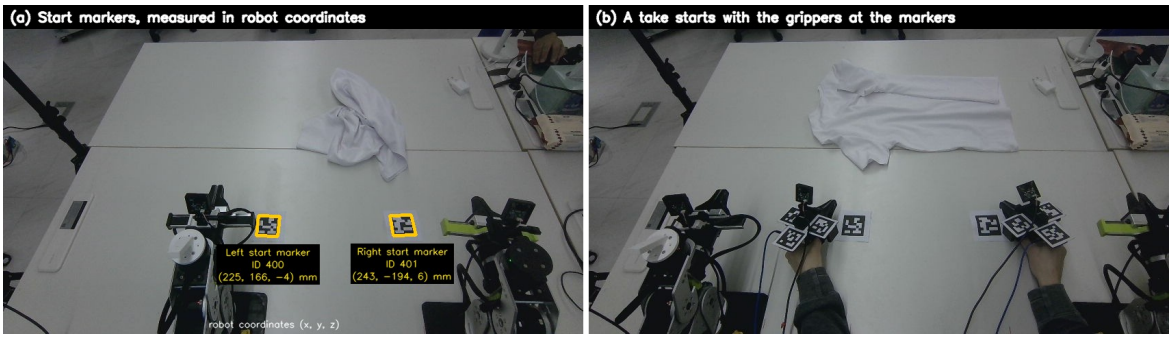


Fig. 4. Start markers. (a) Two AprilTags taped on the table (IDs 400 left, 401 right), measured once with the calibrated fixed camera; their centres in robot coordinates are shown. (b) Every take starts with the two grippers at these markers, so each demonstration begins at known robot coordinates.

VI. Pose of the Hand-Worn Grippers

A. Markers

In each fixed-camera frame the AprilTags are detected and each is solved with IPPE; tags with a side under 18 px are ignored. The marker distance, the least accurate PnP coordinate at about 1 m, is corrected along the viewing ray by the aligned depth when the median depth inside the tag is within 0.10 m of the PnP value. With two or more tags of one rig visible, all corners are solved jointly (SQPnP, then Levenberg–Marquardt), and frames with a reprojection RMSE above 2 px are dropped. The rig pose, the jaw-tip offset and T_{BC} then give the jaw-tip pose in B directly.

B. Point tracking anchored by markers (variant Cg)

Markers are hidden whenever the cloth, the palm or the other hand covers them, which is exactly at grasps. The position therefore comes from point tracking on the fixed camera, with the markers as anchors:

1) *Points.* CoTracker3 (online mode) follows 40 query points per gripper through the colour video, seeded on the gripper body at the start of each demo.
2) *3-D.* The aligned depth lifts the points to 3-D; a rigid gripper model fitted to them gives the gripper body per frame.
3) *Gates.* A frame is invalid when the track acceleration disagrees with the IMU acceleration (normalised cross-correlation < 0.2 over 1 s windows, unless visible tags agree with the track), when the two hands come closer than 8 cm, or when visible tags are more than 10 cm from the track on at least 3 frames within 1 s.
4) *Fusion.* One sparse least-squares problem per demo on a 20 Hz grid with Cauchy weights: the valid track points, the tags where visible, a constant 3-D bias, a second-difference smoothness prior, and IMU acceleration *only* on nodes without other support. Rotation comes from a Mahony filter with the gyro bias taken from rest periods and the yaw anchored at the first frame with at least two tags.

Accuracy. Accuracy is measured by hiding markers that were actually visible and comparing the estimate with them; nothing in a hold-out run (yaw anchor, tag factors, bias) is estimated from the hidden tags. On held-out takes, not used for tuning, the hidden-marker error has a median of **16 mm (left) and 19 mm (right)**; markers with IMU interpolation alone give about 42 mm. On a development take the markers reduced the absolute error from 42 to 12 mm, and adding IMU acceleration everywhere made the result worse, which is why the IMU is used only inside gaps. In one GOLD flattening demo the left hand had markers in 48 % of frames, point tracking only in 49 % and IMU in 3 %; the right hand 77 / 23 / 0 %. Every demo gets a QA video (Fig. 5).

C. What we tried before, and why it failed

Before Cg, three pose sources were tried. (i) *Wrist-camera SLAM.* MASt3R-SLAM on the wrist fisheye, with automatic restarts after tracking loss, was accurate where it tracked (2–14 mm on held-out marker frames) but kept only 19–48 % of frames and 0–38 % of the marker-hidden frames around grasps, and joins between restarted segments jumped 40–73 mm: white cloth fills the view at close range and the hand moves fast. ORB-SLAM3 mono-inertial initialised with good rotation, but its translation scale diverged by 30–1000×; hand accelerations (median 0.07–0.16 m/s^2) were below the accelerometer bias (0.25–0.43 m/s^2), so scale was not observable. Stereo VIO with an earlier stereo camera diverged, and that camera was dropped on 16 October. (ii) *An earlier smoother* (SLAM + external body tracking + IMU) passed all its own quality checks yet was 4–6 cm from the jaw position seen by the external camera, because of a stale camera–IMU rotation (17–24° off), a lever arm that converged 20–35 cm wrong, and a wrong anchor; internal residuals could not see them. This motivated the marker rig. (iii) *Markers with IMU fill*, used for the replay track, leave the grasp phases interpolated (about a fifth of the right-hand frames of the replayed demo). The code of (i) and (ii) is in `archive/`.

VII. Jaw Labels

From the wrist video (demonstrations up to 28 October). An ImageNet-pretrained ResNet-18 [25] per hand classifies a fixed crop of the wrist fisheye that shows the jaws, with hysteresis on its output. Since 22 October a YOLO-pose keypoint model finds the two jaw tips, and their distance gives a continuous aperture g (0 closed, 1 open); it agrees with hand

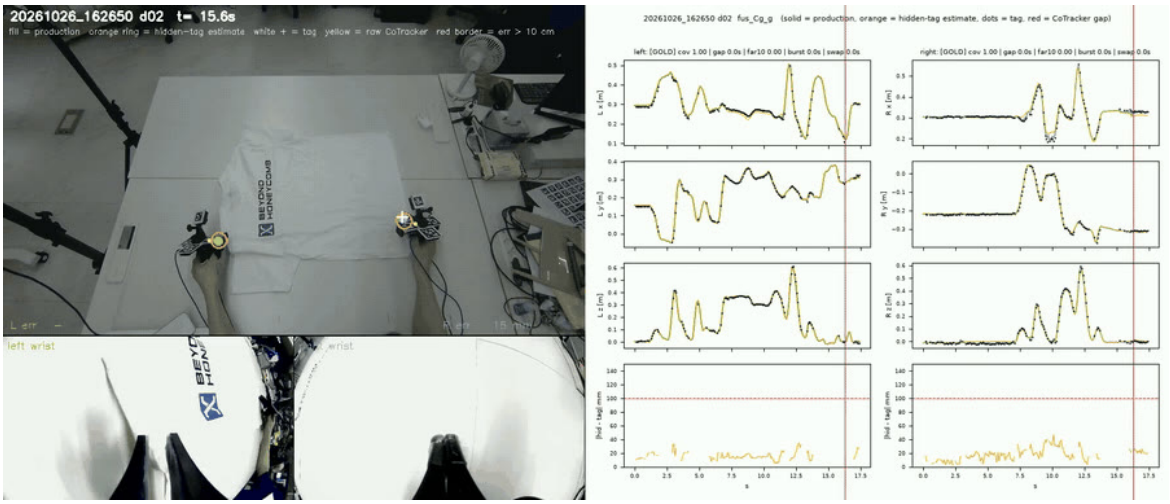


Fig. 5. Per-demo QA video of a GOLD flattening demo. Top left: fixed camera with the fused jaw tips (filled dots), the estimate made with that marker hidden (orange ring), the marker itself (white +) and raw CoTracker points (yellow). Bottom left: both wrist cameras. Right: x, y, z of each hand over time and the hidden-marker error (bottom row; red dashed line = 10 cm).

labels on 100 % (left) and 98 % (right) of frames of a re-angled take.

From the servo (demonstrations since 29 October). The torque-off servo is read at about 100 Hz on the same host clock as the cameras. Its position is mapped through a per-hand calibration (closed … open ticks) to g, then to open/close events with hysteresis and a minimum dwell. The reading is re-centred per recording, because unplugging or power-cycling a servo shifts its whole range; the recorder warns aloud when a servo reads outside its calibrated range. The output has the same format as the video labels, so every later stage is unchanged.

Never mixed. On recordings that have both, the video classifier agrees with the servo on only about 70–80 % of frames (per frame, on the servo-equipped takes). Video- and servo-derived labels are therefore kept apart and never mixed in one comparison or training set. A model trained to reproduce the servo from the wrist video, for relabelling older demonstrations, was evaluated; a candidate dataset built with it on 31 October has not been used for any reported model.

VIII. Replay Track: Converted Demonstrations on the Robot

The replay track (8–22 October) converted one demonstration into a robot trajectory and replayed it open loop. It used the marker pose with IMU fill and video jaw labels, and is kept here because its measurements are the evidence that the calibration and conversion are geometrically sound.

A. Conversion and execution

Robotization changes the human path only by continuous warps, never by clipping: a lateral compression toward each arm's own side with a softplus ceiling at the midline (minimum separation 0.06 m) and a softplus floor above the table; every frame stores its displacement and a quality class (thresholds 20 and 50 mm). *Planning* uses joint-limit-aware differential IK (Pink) with position prioritised over orientation, a bound of 5° per frame on each joint, wrist-roll relaxation about the tool axis, the jaw's 180° symmetry, and a finger-corner clearance model; jaw commands are issued early by the measured jaw latency (open 320 ms, close 480 ms). *Safety checks* run offline before anything moves: IK error, joint limits and speed, TCP and link heights above the table, and at least 120 mm between the links of the two arms. Commands stream at 30 Hz to the 100 Hz bridge; at 0.35× speed a 33 s demo takes 94 s.

Metrics. Three levels are compared in B on the host clock: A the human jaw tip, B the robot command (FK of the commanded joints) and C the robot (FK of the encoders). A is taken relative to the start of the window, B and C relative to their first sample, and C is shifted by the lag in 0–1.5 s that best matches B, so a pure delay is not error. We report position RMSE and p95, the geodesic rotation error, the amplitude ratio (C over A), C−B (tracking) and B−A (what conversion and planning changed). Single-arm gates: RMSE <20 mm, p95 <30 mm, rotation median <5° and p95 <10°, amplitude ratio 0.9–1.1. *Relative metrics hide constant offsets*; absolute placement was checked by tape and video.

B. Results

Single-arm parity. Six-direction translation strokes: 16/18 pass, median per-axis RMSE 5.5–11.6 mm (max 14.5 mm), endpoint repeatability ≤8 mm; one failure was a marginal cross-axis leak, the other a poor human stroke. Rotation strokes: amplitude ratio 0.98–1.04, median error 0.3–1.6°. Free combined motion near human speed: 2/2 pass (RMSE 13.7 / 14.6 mm, p95 26 / 29 mm, rotation median 1.7 / 1.4°, robot lag about 0.2 s). B−A was at most 3.4 mm, so the residual was robot tracking, not conversion. The robot jaw command fell within 0.5 % of window progress of the human jaw event.

Controller finding. The moving error did not change when the proportional gain was raised. With an integral-only feedforward $\tau_{\mathrm{ff}}[k{+}1] = \tau_{\mathrm{ff}}[k] + K_i e[k]\Delta t$ and no gravity model,

TABLE III
TWO-ARM REPLAY, RANGE OVER FOUR RUNS WITH ONE PLAN (WHOLE DEMO AS ONE WINDOW; REFERENCE = ROBOTIZED PATH)

	right	left
position RMSE	17.2–17.4 mm	29.0–30.3 mm
position p95	35–37 mm	50–51 mm
rotation median	8.1°	12.3–12.7°
rotation p95	12.6–12.7°	17.0–17.2°
C−B (tracking)	12.1–12.3 mm	14.5–15.5 mm
B−A (conversion, IK)	8.6 mm	22.6 mm

a quasi-static analysis of $\tau_g = K_p e + \tau_{\text{ff}}$ gives a moving error $e \approx \dot{\tau}_g / K_i$, independent of K_p; a simulation explained 77 % of the measured shoulder-joint error. Raising K_i in two steps reduced the forward-stroke RMSE from 25–37 mm to 12–14 mm (earlier settings had passed 0/6, 3/6 and 5/6 strokes).

First cloth grasp. After the TCP correction, one arm replayed the grasp window of a demonstration and picked up and lifted the T-shirt. The first attempt scraped the table at a strongly tilted tool, which led to the finger-corner clearance model.

Two-arm replay of a full demonstration. One 33 s demonstration was replayed at about a third of human speed (94 s; Fig. 6). In its window the right hand had a visible marker in 79 % of frames and the left in 95 %; 21 % of the right-hand frames were interpolated. The robotization displacement had a p95 of 9.3 mm (right) and 20.6 mm (left). The plan kept the IK error within 6.3 mm/ 11.5° (right) and 10.3 mm/ 21° (left), and the arms stayed at least 27 cm apart. Table III gives the error of four runs with one unchanged plan, against the robotized path that was replayed. The runs agree within about 1 mm: the robot is deterministic and the remaining error is systematic. Against the raw human path the RMSE is about 22 mm (right) and 32–34 mm (left), because the robotization displacement then also counts as error. The left command already deviates more from the replayed path (B−A 22.6 mm vs 8.6 mm): the left IK gives up more orientation.

Jaw height at grasp. With a conservative finger-corner clearance the jaws closed 19–61 mm above the table and missed the cloth. Allowing the finger corner to touch the table plane, as human fingers do, brought the closes to 14–16 mm (right, 4 closes) and 5–19 mm (left, 5 closes) above the table, within 2–8 mm in xy of the command at that moment; the human closes at about 10 mm (Fig. 7). Tilting the tool toward vertical instead failed (the IK could not follow even 15°; errors up to 100 mm), and closing at the lowest nearby point, a heuristic, was rejected.

Outcome. In two runs the shirt lay 15–20 cm farther away than in the demonstration and the jaws closed on the empty table. In one run the jaws caught the shirt edge and dragged it; the shirt was moved, but not folded as by the human. Open-loop replay needs the cloth where the human's was; the next step is therefore a policy that sees the cloth.

IX. POLICY TRAINING DATA

Collection. 318 demonstrations of *flattening* a crumpled T-shirt were recorded on 26 and 27 October; 300 were successful (beige 72, black 92, white 82, blue 54), judged on the frame 0.5 s before each demo ends. The camera sits at 113 cm so that both hands stay in view. Each gripper track is assigned to an arm by its side of the table, because the rig labels on the 22 October and later takes were found swapped against the wrist cameras.

Trims. The waiting start and the still end would be learned as actions. Each demo is cut to 0.3 s before the first and 0.3 s after the last moment either hand moves faster than 10 cm/s. A one-hand tracking gap of at least 0.8 s that starts within 0.5 s of the window start (or ends within 0.5 s of its end) is cut away instead of grading the demo down, unless less than 8 s would remain.

Grades. A demo is **GOLD** if both arms have coverage $\geq$0.95, no gap over 1 s, at most 0.05 of valid frames more than 10 cm from a visible marker, no such burst over 0.5 s, no hand-swap frames, a clean first second and no gap of 1 s or more within $\pm$1 s of a jaw event; **REJECT** if any arm has a gap over 3 s, coverage below 0.80, more than 0.20 of frames far off, a burst over 2 s or swaps over 1 s; otherwise **SILVER**. Whole demos are kept or dropped. Of the first 139 graded demos, 133 are GOLD, 5 SILVER and 1 REJECT. The flattening fine-tunes use 281 GOLD demos from all 300; the final grade counts over all 300 are not reported here.

Labels before IK. The earlier label B was the IK→FK of the robotized path. IK makes execution choices (wrist-roll relaxation, a 180° tool flip near joint limits, pushes that keep the arms apart) and bakes them into the data, and it moved the target most exactly where it matters: on the first 139 demos (about 123k arm-frames) IK→FK moves the target by p95 $\approx$53 mm overall and $\approx$58 mm at grasps (Table IV). On the same demos 20.3 % of frames need a large IK correction, mostly from execution choices, and 0.8 % are truly out of reach (hands lifted far above the table). The label is therefore the Cartesian jaw-tip pose before IK, stored absolute in B; IK is a checker offline and the executor online, and B is kept for QA only. With labels F and A the IK result changes no value and masks no frame; a demo is dropped only if IK cannot be run on it at all.

Label F: no clamp (29 October). Label A was the fused jaw-tip pose in reBot gripper axes, smoothed over 0.25 s, plus a table-clearance step (tilt toward vertical up to 30°, then a lift of the finger corner). That step lifted 44 % of label frames, by 27 mm at the median (max 65 mm), so a policy could not learn to go down to the cloth (Sec. X). The current label F is the same pose without the clearance step: smoothed over 0.25 s, no clamp, lift or feasibility edit; IK does not change or mask it. Limits belong to deployment, not to training labels.

Time stretch. Human demonstrations are faster than the arms can follow (jaw tip p95 0.93 m/s, 127°/s). Time-stretched copies of the data set resample pose, jaw events and every camera frame (nearest source frame) on one common slower

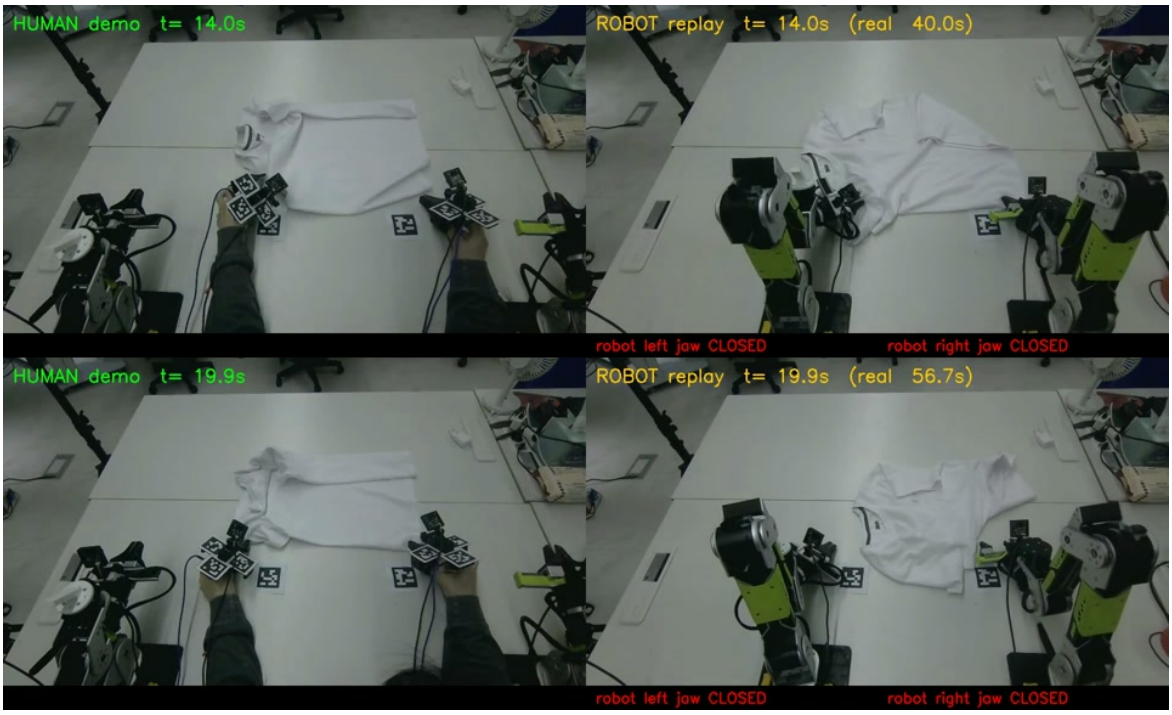


Fig. 6. Human demonstration (left) and two-arm robot replay of the converted trajectory (right), filmed by the same fixed camera and aligned on one clock (robot at 0.35×). The bars show the robot jaw commands. The jaws hold the shirt edge and drag it; the shirt is moved, but not folded as by the human.

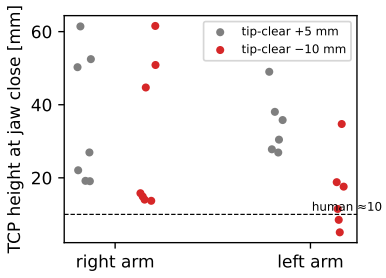


Fig. 7. TCP height at each jaw close with the conservative finger-corner clearance and with the corner allowed to touch the table plane. Dashed line: the human (≈10 mm).

TABLE IV
IK CHECK OF THE CARTESIAN LABEL (FIRST 139 DEMOS). "LARGE" = IK POSITION ERROR >30 MM OR ROTATION >30°; CAUSES ASSIGNED IN ORDER, FIRST MATCH WINS.

	all	left	right
IK→FK vs Cartesian, position p95	53 mm	70 mm	28 mm
frames with a large correction	20.3 %	27.8 %	12.9 %
wrist-roll relaxation	7.6 %	14.8 %	0.4 %
other (IK lag)	4.1 %	5.9 %	2.3 %
joint limit	3.2 %	2.9 %	3.5 %
tool flip	3.1 %	1.3 %	4.9 %
workspace / table	0.8 %	0.9 %	0.7 %
out of reach	0.8 %	1.3 %	0.4 %
arm spacing	0.6 %	0.6 %	0.6 %

timeline: with a factor s, rows stay at 20 Hz but advance $s{\cdot}0.05$ s of source time, so a demo becomes $1/s$ times longer: 0.3× for the flattening set, 0.5× for the pinching sets.

Dataset. LeRobot v3 at 20 Hz with three cameras (fixed camera and both wrist cameras), a 30-D state (per-arm pose relative to the episode start, both jaws, the left–right relative pose) and a 20-D absolute action (per arm position and 6-D rotation, then both jaws; the jaw target is the binary jaw state with a 0.1 s ramp) with a per-dimension mask. The policy learns relative motion $T_t^{-1}T_{t+k}$, $k = 1 \ldots 32$, built in the training loader.

X. First Policies on the Robot

Deploy loop. Each cycle takes fresh images and the measured joints; FK gives the current TCP; the policy predicts 32 relative steps; the newest TCP re-anchors the chunk (latency); a limiter bounds speed and the table floor; Pink IK solves the joints; safety checks (workspace, link-to-link capsules, table, joint limits) must pass; only the first steps of each plan are committed before re-planning (4 by default for the DiT; Fig. 8). On 27 October a bridge restart without the jaw-opening limit let one jaw open to the end of its travel and break its holder; deployment now refuses to send commands unless the bridge reports the jaw limits.

Policies. (1) A diffusion-transformer policy (DiT, CLIP [27] text, 2 observation frames, horizon 33) pretrained for 300k steps on the public Soft-Fold cloth-folding data ran on the robot in dry runs without safety violations. The DiT fine-tunes on our data start from a DiT pretrained for 100k steps

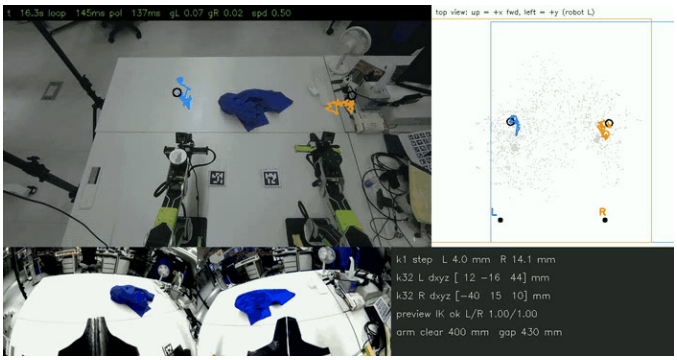


Fig. 8. Deploy UI in a dry run (no command sent): predicted 32-step jaw-tip paths on the fixed camera (left blue, right orange), top view in the robot base, both robot wrist cameras and the live checks.

TABLE V
WHAT THE FIRST POLICIES EXPOSED IN THE DATA (28 OCTOBER – 1 NOVEMBER)

finding	change
Jaw closed early and never re-opened: the policy copied the jaw state from its observation input	jaw state masked to 0 in the observation, in training and deploy
Table clearance in the label lifted 44 % of frames by 27 mm (median)	label F: fused jaw tip, no clamp (Sec. IX)
Demos faster than the arms (p95 0.93 m/s, 127°/s)	time-stretched copies, 0.3× / 0.5×
Video jaw labels disagree with the true aperture	servo encoder labels since 29 October; never mixed (Sec. VII)
Robot wrist cameras tilted 11–13° less toward the tool axis and 25–44 mm closer than the hand-worn ones (table-tag PnP)	re-mount to the hand-worn geometry (next)

on a 1 : 1 : 1 mix of Soft-Fold, iPhUMI [29] and PrimeBot UMI [30]. (2) The X-VLA base model [28] pretrained on a 1 : 1 : 1 mix of the same three datasets, one domain prompt per source. Both were fine-tuned on 281 GOLD flattening demos (from late October) and run with the receding-horizon loop. What is known from these runs is qualitative: the arms approach the cloth but stop short of it and close the jaws early. The trials were not counted, and for the runs of 28–31 October the checkpoints, label kinds and speeds were not recorded. **No success rate is reported.**

Causes and data problems. Two causes were recorded from the robot runs: the jaw-state copying and the wrist-camera placement (first and last rows of Table V). The other three rows were found afterwards by checking the labels. All but the camera re-mount are changed; re-mounting the robot wrist cameras is next.

Also added and *not yet validated on the robot*: a real-time-chunking blend between plans (the UI default since 29 October; whether it was on in a given robot run was not recorded) and an X-VLA policy server.

TABLE VI
PINCHING DATA, STATUS OF 1 NOVEMBER (KEPT = GOLD OR SILVER; BUILT = EPISODES IN THE DATASET)

recorded	takes	demos	kept	built
30 October	5	74	53	53
1 November, T1 + T2	2	60	56	55
1 November, T3	1	35	33	33
1 November, T4	1	10	10	10
1 November, T5	1	50	48	46

XI. PINCHING FEASIBILITY TEST (FIRST ROBOT GRASPS)

To separate "the pipeline cannot teach a grasp" from "flattening is too hard for this amount of data", the first robot feasibility test was reduced to one sub-skill: **pinch one sleeve with the left hand**. The right arm is idle in all data; in the robot runs of 2 November both arms were under policy control and the right arm stayed at its start pose.

Data. All sets use the Cg trajectory source (Fig. 9) with label F, servo jaw labels and the 0.5× time stretch, and keep GOLD and SILVER demos (Table VI). An earlier trajectory source (A′, one position bias per demo and arm from the tags of the first 2 s) had NaN spans inside some demos of 1 November, and IK failed on 4 of them, so every set was rebuilt from Cg; the two sources are never mixed. On 1 November the gate kept 147 demos; the dataset converter dropped 3 whose trimmed window was shorter than 5 s, so with the 53 of 30 October the training set has 197 demonstrations (53 + 144; 62,160 frames).

Test sets (never trained on). P20: 19 demonstrations (6207 frames) from two held-out takes, for arm and jaw: all 10 demos of the 30 October take, not gate-filtered, and 9 of the 10 of the 1 November take (the tenth was rejected). A reject set: every 1 November demo the quality gate rejected, used for jaw timing only, because its servo jaw label is valid and its arm label is not.

Model. The DiT fine-tuned on the 53 demos of 30 October (trajectory source A′, label F) is the starting point and the baseline; on P20 built from Cg it scores 19.89 mm (`val_score_mm`: the mean, over $k = 16$ and $k = 32$ and over the left and right arm, of the translation deviation, position only; 19.13 mm on P20 built from source A′). Its score on the 10 demos of P10 had plateaued at 17.1–17.7 mm since about 150 epochs. A fine-tune from these weights on the merged 197-demo Cg set, with a new optimizer and a cosine schedule over 500 epochs, a checkpoint every 25 epochs, gradient clipping off, a jaw BCE head and new train-time image augmentation, started on 1 November (about 4.7 days planned). On P20 its checkpoints after 25 and 50 epochs score 17.29 and 18.89 mm (baseline 19.89 mm); the 50-epoch checkpoint is the one that grasped on the robot, so the offline score did not pick the robot checkpoint. Earlier in this chain, a DiT pretrained on the three-dataset mix was fine-tuned on the flattening GOLD demos (label F, 0.3×) and then on pinching data, before the 30 October set.

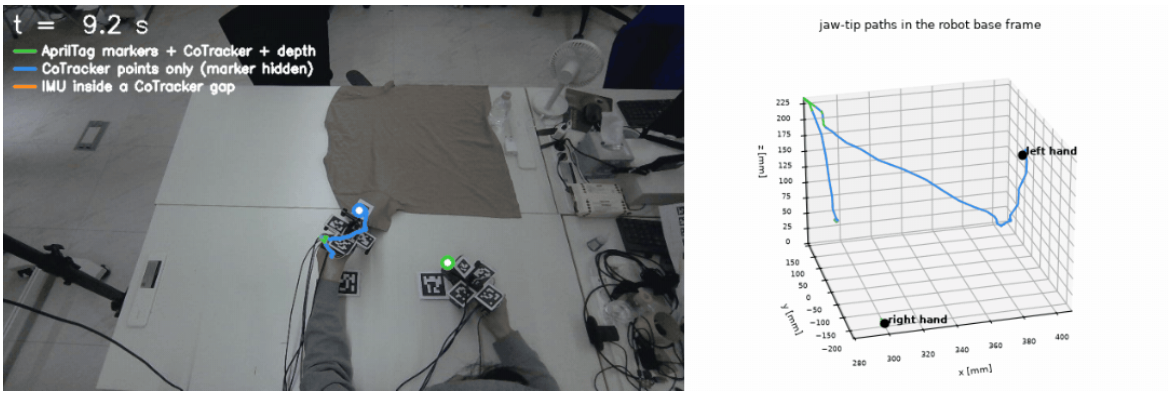


Fig. 9. A GOLD pinching demonstration recorded on 1 November: Cg jaw-tip paths coloured by the source of each frame (green: a gripper marker visible; blue: point tracking only, marker hidden; orange: IMU inside a tracking gap). Left: projected into the fixed camera; right: both hands in the robot base frame. The right hand stays at its start marker.

Robot-side changes. Per-arm table-height calibration from jaw-tip touches on the table tags (plane tilt 4.9° left, 2.6° right), added to every committed command when the UI option is on; the fixed camera, which had moved since 26 October, re-aimed to its earlier view (residual about 5 px / 0.4°); and a soft stop that brakes on the current command without latching an emergency stop.

Robot runs of 1 November (qualitative, not counted). Two DiT checkpoints were run with 32 steps committed per plan and speed 1 (later 1.5): the flattening fine-tune on 281 GOLD + 48 servo-labelled demos (label F, 0.3×; two runs of about 70 s) and the pinching baseline (one short run). In both the left jaw was mostly commanded open and the right jaw stayed closed; in the first run the left wrist had one motor fault and the left arm once lagged its command by more than 20°. No success or failure was scored.

Robot runs of 2 November (first grasps; trials were not counted). The 50-epoch checkpoint of the 197-demo fine-tune was run with 32 steps committed per plan, speed 1, real-time-chunking look-ahead off, 10 DDIM sampling steps, per-arm height calibration on and the soft stop. Successful grasps of the sleeve were observed in real-robot trials on 2 November (Fig. 10): the left arm approaches the sleeve end with the jaw open, closes on it and pulls it. The trials were not counted, so no success rate is given. After a miss the policy grasps elsewhere instead of retrying: the data contain no recovery demonstrations, so post-failure states are out of distribution. A retry means sending the arms back to the start pose and starting again; the data fix is to record recovery demonstrations. Hardware changes that day: the arms' USB locations had swapped after re-cabling (fixed in the launch configuration); the left jaw has mechanical slack and now closes 25° past its zero, with the extra squeeze ramped in over the last 30° of travel; the start move now closes both jaws (a parked jaw relaxes open); one robot wrist camera stalled once and the deploy UI was restarted.

Status: first grasps observed, trials were not counted. Robot trials will be reported with the number of trials and successes, whatever the outcome.

XII. Flattening after the Pinching Test (2–5 November)

After the first pinching grasps the work returned to flattening, with both hands, Cg trajectories and servo jaw labels. The numbers of this section are in `docs/status_2026-11-05.md`.

FLAT149: the UMI recipe. 149 flattening demonstrations (48 of 30 October and 101 of 2 November; a demo was kept only if the shirt ended flat in the fixed-camera frame) were built at 0.5× speed with a continuous jaw target. Training followed the cloth-folding recipe of the UMI appendix [1] item by item: 100 epochs, learning rate $3 \cdot 10^{-4}$ (vision encoder $3 \cdot 10^{-5}$), DDIM with 50 training and 16 inference steps, and image augmentation matching UMI's random crop (zoom 0.05, shift 0.025), rotation ($\pm 5°$) and colour jitter (brightness 0.3, contrast 0.4, saturation 0.5, hue 0.08). The jaw is learned as a continuous value (no BCE head) and binarised at deployment. Training started from the H100 checkpoint of the GOLD-281 flattening fine-tune and ran on four H100s. The hue shift is a rotation of the chroma plane in YIQ space (one 3×3 matrix): the library hue adjustment cost about 62 ms per camera sample on the CPU data workers and made the run data-bound at about 3 s per update. On the held-out sets the best validation checkpoint is at epoch 80 (`val_score_mm` 60.89) and the best test checkpoint at epoch 90 (64.36). No counted robot result exists.

Deploy changes (3 November). A motion-amplitude scale (`amp`) shrinks each committed step's translation and rotation about the current pose (the user found the motion too large); the start move can use the medoid start of any dataset (`start_root`; the FLAT149 starts lie 19 / 31 mm (left / right) from those of GOLD-281); a camera-alignment tool compares the live view with a reference day.

Camera re-aimed, new collection, split by tracking gaps (3–4 November). The fixed camera was re-aimed during a take on 3 November; that take and later ones use a re-solved calibration from the two table anchor tags and the table plane (the two coplanar tags make the PnP initialisation ambiguous,

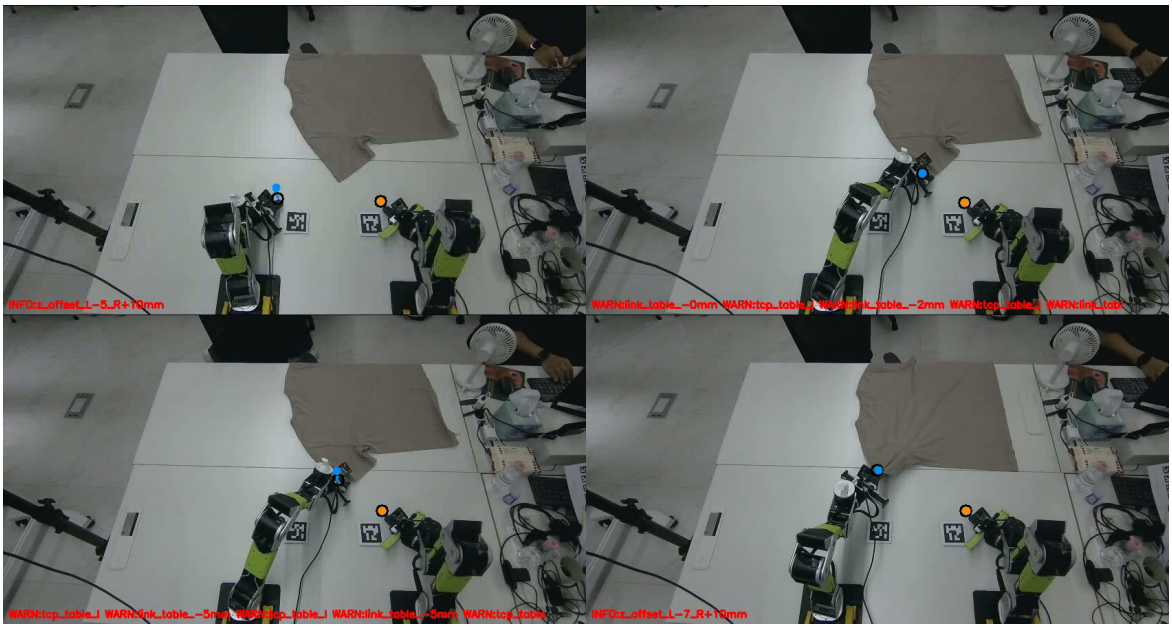


Fig. 10. One of the real-robot runs of 2 November judged successful by the operator (50-epoch checkpoint; fixed camera, one frame per plan cycle; trials were not counted). Top left: start; top right: the left arm approaches the sleeve end with the jaw open (7 s); bottom left: jaw closed on the sleeve (11 s); bottom right: the sleeve pulled (22 s). Blue / orange dots: the policy's next targets for the left / right jaw. Red text: the deploy loop's table-proximity messages.

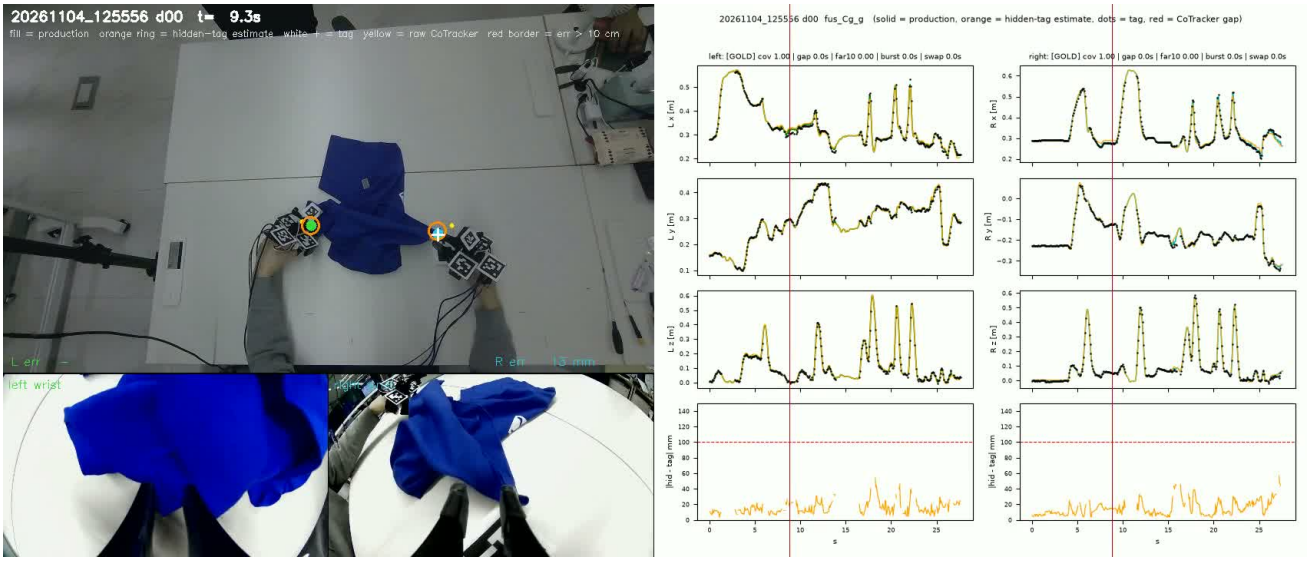


Fig. 11. Per-demo QA video of a GOLD flattening demo under the re-aimed camera (4 November, train split). Top left: fixed camera with the fused jaw tips (filled dots), the estimate made with that marker hidden (orange ring), the marker itself (white +) and raw CoTracker points (yellow). Bottom left: both wrist cameras. Right: x, y, z of each hand over time and the hidden-marker error (bottom row).

so every IPPE solution is tried and the one with the camera highest above the table is kept), and are kept apart from earlier data. On 3 November 7 takes with 100 demonstrations were recorded (Fig. 11). Every demo is trimmed at both ends (both arms slower than 3 cm/s for 0.5 s, margin 0.3 s), and the longest point-tracking gap is measured *inside* the trimmed window: **train** = GOLD with a longest gap below 0.25 s; GOLD with a longer gap alternates between **validation** and **test**; SILVER without any gap also goes to train, other SILVER and REJECT form separate sets for auxiliary evaluation and analysis. Datasets are built at 0.3× with label F and the continuous jaw; timelines and images are never cut. Positions inside a tracking or marker gap of at least 0.3 s are masked in the loss, and once one future step of a predicted chunk is masked, every later step of that chunk is masked too, because the path across a gap is uncertain.

FLAT250 (4–5 November). The per-take sets of 3 and 4 November were aggregated into 250 training demonstrations (444217 frames), 39 for validation and 38 for test, and trained

TABLE VII
FLAT250: OFFLINE SCORE (VAL_SCORE_MM) PER EPOCH, MM, LOWER IS BETTER (STRIDE 16)

epoch	validation	test
0 (initial checkpoint)	71.67	72.27
1	102.36	101.78
4	82.81	82.42
8	60.74	59.85
10	57.64	56.95
12	53.56	54.16
13	52.56	52.32

with the FLAT149 recipe (global batch 1024, 100 epochs) on two RTX 4090 GPUs with distributed data parallel, saving a checkpoint every epoch. Every epoch is evaluated offline on both held-out sets (Table VII): the score first rises, then falls below the initial checkpoint from epoch 6 on (validation). The run was stopped after epoch 13, its best checkpoint, to free the GPUs for the run below; epochs 8 and 10–13 were copied to the robot computer as rollout candidates; no counted robot result exists.

Does the closed jaw hold the cloth? (5 November). The jaw servo is read with torque off, position only: it has no load or current signal, and the closed position does not separate cloth from nothing. The cue therefore comes from the wrist camera. Jaw close events of 343 flattening demos were segmented with a tighter hysteresis for this purpose (which uncovered 148 partial re-opens hidden in the earlier segmentation); the user labeled 585 events blind to any prediction over four rounds, of which 543 map onto the new events (372 held, 171 empty). A ResNet-18 on a crop around the jaw tips of the wrist image reaches, in take-grouped 5-fold cross-validation, per-event F1 0.90 for held and 0.78 for empty. A causal per-frame probability $p_{\mathrm{grasp}}(t)$ at 15 Hz (mean over the last 0.4 s, restarted at every jaw state change) has a median of 0.99 on held and 0.03 on empty events; 12 % of held events dip below 0.2 and 17 % of empty events touch 0.9 at least once. A frozen generator writes p_{grasp} for every frame of a dataset. A new run with the same recipe feeds it into the (otherwise masked) jaw slots of the observation state, on the data of 3, 4 and 5 November (301 training, 46 validation and 45 test demonstrations); after two epochs its offline score is 68.79 / 68.43 (validation / test; other held-out sets than FLAT250). The deploy loop does not compute p_{grasp} yet.

XIII. SUPERSEDED APPROACHES

Table VIII lists what was replaced, when the current pipeline was built, and why. The code of dropped approaches is in `archive/`, with a README table of dates and reasons.

XIV. LESSONS

1) *Markers plus a fixed camera beat wrist SLAM for contact-rich hand motion.* The failure regime of SLAM (close range, textureless cloth, fast motion) is exactly the grasp.
2) *Anchor, do not replace.* Point tracking covers the hidden phases, and the markers remove its absolute error; neither is enough alone.
3) *Measure in the world, not in the model.* The 89 mm TCP error and the 70 mm base error were invisible to FK-based checks; tape, camera and cloth found them. The 4–6 cm error of the earlier smoother was invisible to its own residuals.
4) *Fix the conversion, do not add heuristics.* Lifting the jaws for clearance cost the grasp; allowing contact restored it.
5) *Do not let the executor, or a safety limit, write the label.* IK choices and a table clamp are right for execution and wrong as training targets.
6) *One clock for everything.* It made camera, IMU, servo and robot alignment, and side-by-side videos, possible without synchronisation hardware.
7) *Model the controller.* The tracking error followed $\dot{\tau}_g / K_i$; changing K_p could not help.

XV. LIMITATIONS

- *No success rate.* Neither replays nor policies were run as counted trials; outcomes are judged by measurement and video.
- *One setup, one demonstrator.* All numbers come from one physical setup and one person; one demonstration was replayed in full on both arms, and its 1 mm repeatability shows determinism, not generality.
- *Relative parity metrics hide offsets.* Absolute placement was checked only by tape at a few poses and by video.
- *Pose accuracy at grasps.* The hidden-marker error is measured where markers can be hidden, mostly away from the grasp; the error inside long tracking gaps is bounded only by the gates.
- *Grades and IK statistics* reported here were computed on the first 139 flattening demos; the final grade counts over all 300 are not reported; success was judged by one person from one frame.
- *Jaw labels* from the wrist video have no held-out accuracy number on the older demonstrations, and agree with the servo on only about 70–80 % of frames where both exist.
- *Offline scores* such as `val_score_mm` measure agreement with held-out human motion, not task success.
- *Grasp labels* come from one labeler; the frozen p_{grasp} generator is trained on all labeled events, so its output on those events is in-sample.
- *Hardware.* The left robot wrist connector has dropped out repeatedly.

XVI. REPRODUCIBILITY

Repository layout. One installable package per pipeline stage, each run as `python -m <package>.<module>`: `src/laundry_tracking/`: camera-to-robot and marker-rig calibration, point tracking, marker and IMU fusion, gates, trims and grades; `src/laundry_jaw/`: jaw labels from the servo and from the wrist video; `src/laundry_dataset/`:

TABLE VIII
CURRENT PIPELINE AND WHAT IT REPLACED

part	current (1 November)	replaced	why
Pose source	CoTracker3 on the fixed camera + marker anchors + constant bias, IMU only in gaps (Cg)	MASt3R wrist-SLAM gap filler; ORB-SLAM3; stereo VIO; SLAM + body-tracking smoother; markers + IMU fill (replay track)	lost at grasps; scale unobservable; 4–6 cm label error
Camera ↔ robot	re-solved after every camera move from two table tags + the table plane	one 32-pose robot-held-marker solve	camera raised so both hands stay in view
Jaw state	servo encoder (since 29 October); video model for older demos	red-tape jaw detector (tape removed 20 October)	detector fired on background; video labels disagree with the servo
Arm assignment	by the gripper's side of the table	by rig label	labels swapped
Label	F: fused jaw-tip pose (gripper axes, 0.25 s smoothing) before IK, no clamp	B: IK→FK; A: with table clearance	IK choices and the clamp moved the target at grasps
Data speed	0.3× / 0.5× time-stretched copies	human speed	too fast for the arms
Execution	receding-horizon deploy, 4 steps per plan (DiT default)	whole-demo open-loop replay	replay needs the same scene

labels, IK diagnostics, the LeRobot dataset and the flattening split and builds; `src/laundry_training/`: the DiT trainer; `src/laundry_deploy/`: receding-horizon deploy, offline evaluation, height calibration and the robot bridge; `src/laundry_shared/`: recorder, jaw classifier, reBot IK, limits, safety, replay planner and shared marker-rig pose code; `src/laundry_common/`: data and checkpoint paths; `scripts/laundry/`: replay robotization and parity scoring, grasp labels and p_{grasp}, cluster jobs; `src/laundry_reference/`: small standalone re-implementations of the trajectory stages that run on the synthetic data in `assets/laundry/examples/`; `configs/laundry/`: calibration results and robot overrides; `hardware/umihand/`: the HandUMI build copy with our modifications (upstream Apache-2.0 licence kept); `archive/`: superseded code; `docs/`, `media/`, `report/`: documentation, media and this report. The mapping from earlier repository paths is in `docs/refactor_2026-11.md`.

Not included. Raw recordings, trained checkpoints, the reBot vendor plugin and URDF, ORB-SLAM3 source, and the X-VLA fine-tuning code. Real-data stages need the recordings, some need a GPU or the robot.

How to verify. From a checkout, `pip install -e ".[test]"` (or `uv sync`), then `python -m pytest -q` (characterization tests of every package, regression of the standalone tools on the synthetic example, import and config-path smoke tests), `python scripts/laundry/eval/replay_parity.py --synthetic` and `python -m laundry_tracking.calib.robot_cam_solve --synthetic` (self-tests of the parity scorer and the camera-to-robot solver on simulated data), all run in CI. `python analysis/verify_claims.py` checks that every number in this report also appears in the in-repository document it was taken from (README, `docs/pipeline.md`, `docs/history.md`, `docs/results.md`, `docs/details.md`, `docs/status_2026-11-01.md`, `docs/status_2026-11-02.md`, `docs/status_2026-11-05.md` or the archived report v0.1). This is a check of consistency with the project's records, not a re-computation of the numbers from data.

XVII. CONCLUSION AND NEXT STEPS

With one world frame and one clock, converted hand-worn-gripper demonstrations replay deterministically on a two-arm robot and bring its jaws to the human's grasp height, which shows that calibration and conversion are geometrically sound; open-loop replay does not fold a shirt. A pose source that anchors fixed-camera point tracking with markers keeps the jaw-tip pose through grasps, and the training labels are now the fused jaw tip before IK, without a clamp, with servo jaw labels and slowed timing. The first policies, observed in uncounted robot trials, stopped short of the cloth; the recorded causes and the label problems found afterwards are changed, except the camera re-mount. In the reduced pinching test, a policy fine-tuned on 197 demonstrations grasped the sleeve in uncounted robot trials on 2 November; it does not recover from a miss. Next: record recovery demonstrations, re-mount the robot wrist cameras, finish the pinching test with counted robot trials, then evaluate the flattening policies trained under the re-aimed camera (FLAT250, and its variant conditioned on p_{grasp}) with counted trials.

ACKNOWLEDGEMENTS

The hand-worn gripper is HandUMI by BrikHMP18 (murobotics-ai, Apache-2.0), following UMI [1]. Pretraining data: Soft-Fold, iPhUMI (MIT), PrimeBot UMI (CC BY-SA 4.0).

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